



Name: _____

Teacher: _____

Class: _____

FORT STREET HIGH SCHOOL

2010

HIGHER SCHOOL CERTIFICATE COURSE

ASSESSMENT TASK 3: TRIAL HSC

Mathematics Extension 1

TIME ALLOWED: 2 HOURS

(PLUS 5 MINUTES READING TIME)

Outcomes Assessed	Questions	Marks
Chooses and applies appropriate mathematical techniques in order to solve problems effectively	1, 2	
Manipulates algebraic expressions to solve problems from topic areas such as inverse functions, trigonometry and polynomials	3, 4, 5	
Uses a variety of methods from calculus to investigate mathematical models of real life situations, such as projectiles, kinematics and growth and decay	6	
Synthesises mathematical solutions to harder problems and communicates them in appropriate form	7	

Question	1	2	3	4	5	6	7	Total	%
Marks	/12	/12	/12	/12	/12	/12	/12	/84	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each new question is to be started in a new booklet

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Question 1 (12 marks)

- a) Differentiate $\cos^{-1}\left(\frac{3x}{2}\right)$ with respect to x . 2
- b) Find the acute angle between the lines $2x + y - 3 = 0$ and $x = 1$. 2
Correct your answer to the nearest degree.
- c) Find the coordinates of the point P which divides the interval AB with endpoints 3
 $A(2, 3)$ and $B(7, -7)$ externally in the ratio 4:9.
- d) Evaluate $\int_0^1 \frac{x+1}{x^2+1} dx$ 3
- e) Find all the real values of a for which $ax^3 - 8x^2 - 9$ is divisible by $(x - a)$ 2

Question 2 **(12 marks)** *Start a separate booklet*

- a) Solve the inequality $\frac{2x-3}{x+2} \geq 3$ **3**
- b) i) Show that a solution for $3\sin x - x = 0$ lies between $x = 2.2$ and $x = 2.4$. **1**
- ii) Taking $x = 2.3$ as an initial approximation for a solution to $3\sin x - x = 0$, **3**
 apply Newton's method once to find a better approximation correct to three decimal places.
- c) Find the values for a for which $f(x) = e^{-ax}(x-a)$ has a stationary point at $x = \frac{5}{2}$. **2**
- d) Use the substitution $x = \log_e u$ to find $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$ **3**

Question 3 **(12 marks)** *Start a separate booklet*

a) Let $f(x) = 2x - x^2$ for $x \leq 1$.

This function has an inverse $f^{-1}(x)$.

i) Sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ on the same diagram. **2**

ii) Find an expression for $f^{-1}(x)$ **2**

iii) Evaluate $f^{-1}\left(\frac{3}{4}\right)$. **1**

b) i) Express $\sin x - \cos x$ in the form $A \sin(x - \alpha)$ where $0 < \alpha < \frac{\pi}{2}$ **1**

ii) Determine $\lim_{x \rightarrow \frac{\pi}{4}} \left[\frac{\sin x - \cos x}{x - \frac{\pi}{4}} \right]$ **2**

c) Use the method of mathematical induction to prove that if $y = xe^x$ then **4**

$$\frac{d^n y}{dx^n} = e^x (x + n), \text{ for all positive integers } n.$$

Question 4 (12 marks) *Start a separate booklet*

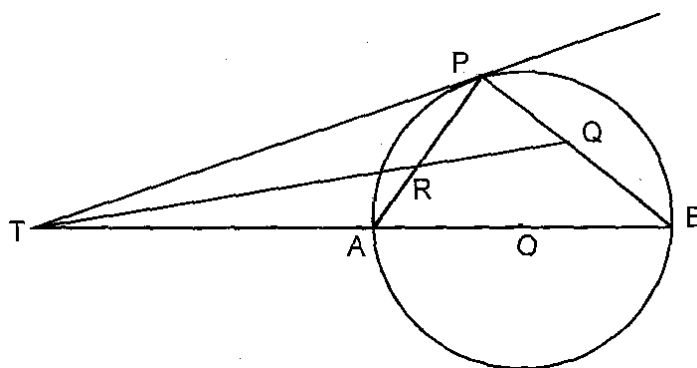
- a) i) A particle moves from the origin with initial velocity $u \text{ ms}^{-1}$ and experiences a retardation of magnitude $v + 2v^2$, where v is the velocity of the particle at time t . 1

Show that when it is at position x , $\frac{dv}{dx} = -(1 + 2v)$

- ii) Find its distance from the origin when it comes to rest. 3

- b) O is the centre of the circle and TQ bisects $\angle OTP$. 4

TB , AP and BP are straight lines and TP is a tangent to the circle at P .



Let $\angle PTQ = \alpha$ and $\angle TBQ = \beta$

Show $\angle TQP = 45^\circ$

- c) $P(2at, at^2)$ is a variable point on the parabola $x^2 = 4ay$ whose focus is S .

$Q(x, y)$ divides the interval from P to S in the ratio $t^2 : 1$.

- i) Find the coordinates of Q in terms of a and t . 1

- ii) Verify that $\frac{y}{x} = t$ 1

- iii) Prove that as P moves on the parabola, Q moves on a circle and state its centre and radius. 2

Question 5 **(12 marks)** *Start a separate booklet*

a) Solve the equation $x^3 - 3x + 2 = 0$, given it has a double root. **2**

b) i) Show that $\cos 3x = 4\cos^3 x - 3\cos x$ **2**

ii) Show that the solution of $\cos 3x - \sin 2x = 0$, for $0 < x < \frac{\pi}{2}$ is given by **3**

$$\sin x = \frac{\sqrt{5}-1}{4}$$

iii) Verify that $x = \frac{\pi}{10}$ is a solution to $\cos 3x = \sin 2x$. **2**

iv) Using the results obtained in parts (ii) and (iii) prove **3**

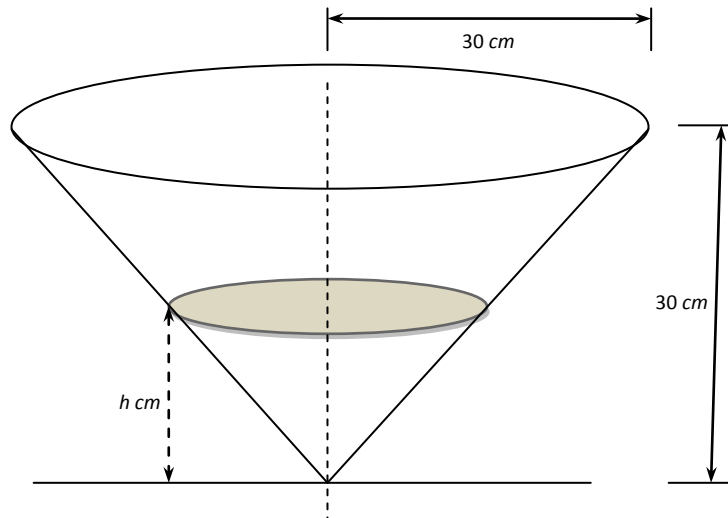
$$\sin \frac{\pi}{5} \cos \frac{\pi}{10} = \frac{\sqrt{5}}{4}$$

Question 6 (12 marks) *Start a separate booklet*

- a) On a certain day the depth of water in a bay at high tide is 11 metres. At low tide, 6.25 hours later, the depth of the water is 7 metres. If high tide is due at 3:20 pm, what is the earliest time at which a ship needing a depth of 10 metres of water can enter the bay? (It may be assumed that the rise and fall of the water level is in simple harmonic motion).

4

b)



4

Water is poured into a conical vessel at a constant rate of 24 cm^3 per second.

The depth of the water is h cm at time t seconds.

What is the rate of increase of the area of the surface of the liquid when the depth is 16 cm?

- c) A particle is projected in a straight line from an origin with velocity 2 ms^{-1} .

When x metres from the origin, its acceleration is $\left(2 - e^{-\frac{x}{2}}\right) \text{ ms}^{-2}$.

- i) Show that, when x metres from the origin, its velocity, $v \text{ ms}^{-1}$, is given by

2

$$v^2 = 4x + 4e^{-\frac{x}{2}}$$

- ii) Explain why, for large positive values of x , $v \approx 2\sqrt{x}$.

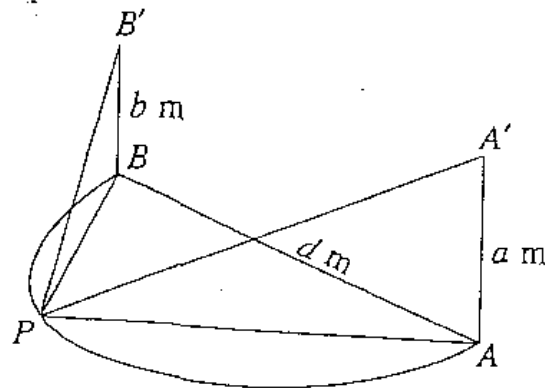
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- iii) Prove that the particle will move from $x = 100$ to $x = 121$ in approximately 1 second.

1

Question 7 (12 marks) *Start a separate booklet*

- a) APB is a horizontal semicircle, diameter d metres. At A and B are vertical posts of height a m and b m. From P , the angle of elevation of the tops of both posts is θ



- i) Prove that $d^2 = \frac{a^2}{\tan^2 \theta} + \frac{b^2}{\tan^2 \theta}$. 2
- ii) From B , the angle of elevation of A' is α and from A , the angle of elevation of B' is β . 2
- Prove that $\tan^2 \alpha + \tan^2 \beta = \tan^2 \theta$.

- b) Two particles are projected at different times from the same point with speed V . The angles of projection of the two particles are α° and $(90 - \alpha)^\circ$ respectively. The greatest heights they reach above the horizontal plane through the point of projection are h_1 and h_2 respectively.

- i) Show that for any angle α , $h_1 + h_2 = \frac{R}{2}$, where R is the maximum range. 4
- ii) If $\tan \alpha = \frac{3}{4}$ and $v = 196 \text{ m/s}$, what time must elapse between the instants of projection if the particles collide as they hit the horizontal plane? (Take $g = 9.8 \text{ ms}^{-2}$). 4

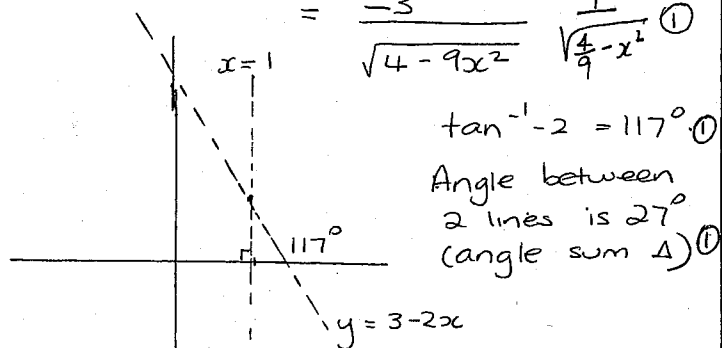
Question 1

SOLUTIONS

$$a) \frac{d}{dx} \cos^{-1} \frac{3x}{2} = \frac{-1}{\sqrt{1 - \frac{9x^2}{4}}} \times \frac{3}{2} \quad (1)$$

$$= \frac{-3}{\sqrt{4 - 9x^2}} \times \frac{1}{\sqrt{\frac{4}{9} - x^2}} \quad (1)$$

b)



$$c) P = \left(\frac{-9 \times 2 + 4 \times 7}{-5}, \frac{-9 \times 3 + 4 \times 7}{-5} \right) \quad (1)$$

$$= (-2, 11) \quad (1)$$

$$d) \int_0^1 \frac{x+1}{x^2+1} dx = \int_0^1 \frac{x}{x^2+1} dx + \int_0^1 \frac{1}{x^2+1} dx$$

$$= \frac{1}{2} \ln(x^2+1) \Big|_0^1 + \tan^{-1} x \Big|_0^1 \quad (1)$$

$$= \frac{1}{2} \ln 2 + \frac{\pi}{4} \quad (1)$$

$$e) P(a) = a^4 - 8a^2 - 9$$

$$= (a^2 - 9)(a^2 + 1) \quad (1)$$

If $(x-a)$ is a factor $P(a) = 0$

$$\therefore a = \pm 3 \quad (1)$$

COMMENTS

IKIAL HSC 2010 : EXTENSION 1

Question 2.

SOLUTIONS

$$a) \frac{(2x-3) \times (x+2)^2}{(x+2)} \geq 3(x+2)^2$$

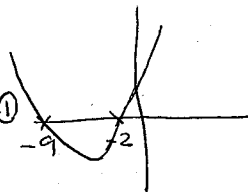
$$(2x-3)(x+2) \geq 3(x+2)^2 \quad (1)$$

$$(x+2)[(2x-3) - 3(x+2)] \geq 0.$$

$$-(x+2)(x+9) \geq 0.$$

$$(x+2)(x+9) \leq 0 \quad (1)$$

$$\therefore -9 \leq x \leq -2 \quad (1)$$



$$b) (i) \text{ When } x = 2.2 \quad 3 \sin x - x = 0.2254 \dots$$

$$x = 2.4 \quad 3 \sin x - x = -0.373$$

Since the sign changes, a solution must lie between these two values. (1)

$$(ii) x_2 = 2.3 - \frac{f(2.3)}{f'(2.3)} \quad (1)$$

$$= 2.3 - \frac{3 \sin(2.3) - 2.3}{3 \cos(2.3) - 1} \quad (1)$$

$$= 2.279 \text{ (to 3 d.p.)} \quad (1)$$

$$c) f'(x) = e^{-ax} \times 1 + (x-a) \times -a e^{-ax}$$

$$= e^{-ax} (1 - ax + a^2)$$

If stationary point exists at $\frac{\pi}{2}$

then

$$f'\left(\frac{\pi}{2}\right) = 0$$

COMMENTS

Mostly well done.

Students should realise that $x \neq -2$

* Everything (values + conclusion) has to be right for 1 mark.

Parts (i) & (ii) were poorly done because

1. Students did not know Newton's Formula (-2 marks)

2. Students do not seem to know how to work with radians ~ many calculator errors +

3. $d(\sin x) = \cos x$ & not $-\cos x$

TRIAL HSC 2010 : EXTENSION I
SOLUTIONS

Question 2 (contd)

$$\begin{aligned} f'\left(\frac{5}{2}\right) &= e^{-\frac{5a}{2}} \left(1 - \frac{5a}{2} + a^2\right) \\ &= \frac{e^{-\frac{5a}{2}}}{2} (2a^2 - 5a + 2) \quad (1) \\ &= 0 \text{ if } \\ &= \frac{e^{-\frac{5a}{2}}}{2} (2a-1)(a-2) \\ &= 0 \text{ if } a = \frac{1}{2} \text{ or } 2. \quad (1) \end{aligned}$$

d) $x = \log_e u$
 $\frac{dx}{du} = \frac{1}{u}$

$$\int \frac{e^x dx}{\sqrt{1-e^{2x}}} = \int \frac{u \times \frac{du}{u}}{\sqrt{1-u^2}} \quad (1)$$

$$= \int \frac{du}{\sqrt{1-u^2}} \quad (1)$$

$$= \sin^{-1} u + c$$

$$= \sin^{-1}(e^x) + c \quad (1)$$

OR

$$x = \ln u \rightarrow u = e^x \therefore du = e^x dx$$

$$\therefore \int \frac{e^x dx}{\sqrt{1-e^{2x}}} = \int \frac{du}{\sqrt{1-u^2}}$$

$$= \sin^{-1} u + c$$

$$= \sin^{-1}(e^x) + c$$

COMMENTS

* Should use product rule to differentiate.

* realise that

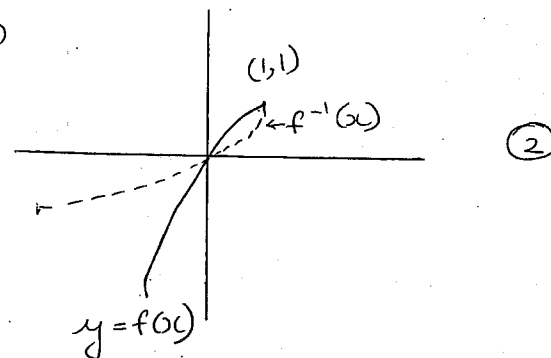
$e^{-\frac{5a}{2}} \neq 0$
 $e^{-\frac{5a}{2}} > 0$ } marks were not deducted if this was not written

TRIAL HSC 2010 : EXTENSION I
SOLUTIONS

Question 3.

COMMENTS

a) (i)



(ii) For $y = 2x - x^2$

inverse is $x = 2y - y^2$

$$x+1 = -(y^2 - 2y + 1) + 1$$

$$x+1 = 1 - (y-1)^2$$

$$(y-1)^2 = 1-x$$

$$y = 1 \pm \sqrt{1-x}$$

From graph we see

$$y = 1 - \sqrt{1-x} \quad (1)$$

$$\begin{aligned} \text{(iii)} \quad f^{-1}\left(\frac{3}{4}\right) &= 1 - \sqrt{1 - \frac{3}{4}} \\ &= \frac{1}{2} \end{aligned} \quad (1)$$

b) (i) $\sin x - \cos x = \sqrt{2} \sin\left(x - \frac{\pi}{4}\right) \quad (1)$

$$\text{(ii)} \quad \lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{\sqrt{2} \sin\left(x - \frac{\pi}{4}\right)}{x - \frac{\pi}{4}} \right) = \sqrt{2} \quad (2)$$

Try to make y subject (1)

$$\begin{aligned} A &= \sqrt{1^2 + 1^2} = \sqrt{2} \\ \alpha &= \tan^{-1}(-1) \\ \alpha &= -\frac{\pi}{4} \end{aligned}$$

← Answer should be given in terms of x .
Lost 1 mark if integration constant was forgotten.

Question 3 (contd) SOLUTIONS

c) $y = x e^x$

Test $n=1$

$$\frac{dy}{dx} = x e^x + e^x \quad \textcircled{1}$$

$$= e^x(x+1) \therefore \text{true for } n=1$$

Assume true for $n=k$

$$\frac{d^k y}{dx^k} = e^x(x+k) \quad \textcircled{1}$$

Consider $n=k+1$

$$\frac{d^{k+1} y}{dx^{k+1}} = e^x + (x+k)e^x$$

$$= e^x(1+x+k)$$

$$= e^x(x+(k+1)) \quad \textcircled{1}$$

This is of the same form as for $n=k$, therefore if true for $n=k$ it is also true for $n=k+1$. Since it is true for $n=1$, it is true for $n=2$ and hence all following $\textcircled{1}$ positive integers.

COMMENTS

Question 4

SOLUTIONS

a) (i) $\frac{dv}{dx} = -\frac{(v+2v^2)}{(1+2v)}$

$$\frac{dv}{dx} = -\frac{(1+2v)}{(1+2v)} \quad \textcircled{1}$$

(ii) $\frac{dx}{dv} = -\frac{1}{1+2v}$

$$x = \int \frac{1}{1+2v} dv$$

$$x = -\frac{1}{2} \ln(1+2v) + C$$

when $x=0$, $v=u$

$$0 = -\frac{1}{2} \ln(1+2u) + C$$

$$\therefore C = \frac{1}{2} \ln(1+2u)$$

$$x = \frac{1}{2} \ln(1+2u) - \frac{1}{2} \ln(1+2v)$$

$$= \frac{1}{2} \ln \left(\frac{1+2u}{1+2v} \right)$$

When comes to rest $v=0$

$$\therefore x = \frac{1}{2} \ln(1+2u)$$

b) $\angle PQR = \angle QPB = \alpha$ (given)

$\angle APB = 90^\circ$ (angle in a semi-circle) $\textcircled{1}$

$\angle PBT = \beta$ given.

$\angle TPA = \beta$ (angle between chord and tangent equals angle in alternate segment) $\textcircled{1}$

$\angle PRQ = \alpha + \beta$ (exterior angle to $\triangle TPR$) $\textcircled{1}$

$\angle PQR = \alpha + \beta$ (" " " $\triangle TQR$) $\textcircled{1}$

In $\triangle PRQ$ both base angles are $(\alpha + \beta)$

$\therefore \triangle PRQ$ is isosceles

$\therefore$ each must be 45°

$\therefore \angle RPQ = 45^\circ$ as req'd. $\textcircled{1}$

COMMENTS

This needed to be demonstrated not just write down.

- many students didn't 'C' & did not use conditions to evaluate 'C'

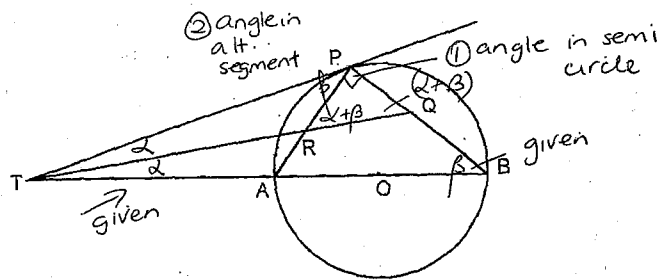
* some students evaluated a definite integral which was fine

→ some students gave this part

o' all poorly done. Many made poor proof unnecessarily lengthy & convoluted

Note student MUST LEARN proper geometric reasoning & state them succinctly.

TRIAL HSC 2010 : EXTENSION I
SOLUTIONS
Question 4 (contd)



c) (i) $P = (2at, at^2)$ $S = (0, a)$ $k:1$
 $x = \frac{bx_2 + dx_1}{k^2 + 1}$ $y = \frac{bx_2 + dy_1}{k^2 + 1}$ $t^2:1$
 $\therefore Q = \left(\frac{2at}{t^2 + 1}, \frac{2at^2}{t^2 + 1} \right)$ ①

(ii) $y = \frac{2at^2}{t^2 + 1}$
 $= \left(\frac{2at}{t^2 + 1} \right) t$
 $= xt.$

$\therefore \frac{y}{x} = t$

At Q
 (iii) $x = \frac{2at}{t^2 + 1}$
 $= \frac{2a \left(\frac{y}{x} \right)}{\left(\frac{y}{x} \right)^2 + 1}$ ①
 $= \frac{2ay}{\frac{y^2 + x^2}{x^2}}$
 $= \frac{2ay \cdot x^2}{x^2 \cdot \frac{y^2 + x^2}{x^2}}$
 $= \frac{2ayx}{1 + \frac{y^2}{x^2}}$

COMMENTS

I was extremely liberal, but you will not be so lucky in the HSC

← learn formula

← many used $\frac{x}{y}$ and I showed it was t .
 (must use L.H.S./R.H.S. setting out)

poorly done.

TRIAL HSC 2010 : EXTENSION I
SOLUTIONS
Question 4 (contd)

$$1 = \frac{2ay}{y^2 + x^2}$$

$$x^2 + y^2 - 2ay = 0.$$

$$x^2 + (y^2 - 2ay + a^2) = a^2$$

$$x^2 + (y - a)^2 = a^2.$$

$\therefore Q$ lies on circle whose centre is $(0, a)$ and radius is a units. ①

COMMENTS

← someone got to here but did not complete the square to find the true centre & radius

Question 5.

SOLUTIONS

a) Roots of $x^3 - 3x + 2 = 0$ are α, α and β .

$$\therefore 2\alpha + \beta = 0 \Rightarrow \beta = -2\alpha$$

$$\alpha^2 + 2\alpha\beta = -3 \quad (1)$$

$$\alpha^2\beta = -2$$

$$\alpha^2x - 2\alpha = -2$$

$$\alpha^3 = 1$$

$$\therefore \alpha = 1 \text{ and } \beta = -2.$$

$\therefore$ Roots are 1, 1 and -2. (1)

b) (i) $\cos 3x = \cos (2x + x)$

$$= \cos 2x \cos x - \sin 2x \sin x$$

$$= (2\cos^2 x - 1)\cos x - 2\sin^2 x \cos x \quad (1)$$

$$= 2\cos^3 x - \cos x - 2\cos x(1 - \cos^2 x)$$

$$= 4\cos^3 x - 3\cos x. \quad (1)$$

(ii) $\cos 3x - \sin 2x = 0 \quad 0 < x < \frac{\pi}{2}$

$$4\cos^3 x - 3\cos x - 2\sin x \cos x = 0.$$

$$\cos x (4\cos^2 x - 3 - 2\sin x) = 0.$$

$$\cos x [4(1 - \sin^2 x) - 3 - 2\sin x] = 0$$

$$\cos x (4\sin^2 x + 2\sin x - 1) = 0. \quad (1)$$

$\cos x = 0$ when $x = \frac{\pi}{2}$ which is not in domain

$$\therefore 4\sin^2 x + 2\sin x - 1 = 0 \text{ ie}$$

$$\sin x = \frac{-2 \pm \sqrt{4+16}}{8}$$

$$= \pm \frac{\sqrt{5}-1}{4} \quad (1)$$

COMMENTS

Many Ext 1 students used factor theorem & polynomial div.

Many Ext 2 students used

$f(x)=0 \quad f'(x)=0$ for double root.

well done

A good number of students ignored the soln to $\cos x = 0$ and lost a mark.

Question 5 (contd)

SOLUTIONS

$\sin x = \frac{\sqrt{5}-1}{4}$ is also outside required range

$$\therefore \sin x = \frac{\sqrt{5}-1}{4} \quad (1)$$

$$(iii) \text{ If } x = \frac{\pi}{10}, \cos 3x = \cos \frac{3\pi}{10} = \sin \left(\frac{\pi}{2} - \frac{3\pi}{10} \right) \quad (1)$$

$$= \sin \frac{2\pi}{10} \quad (1)$$

$\therefore x = \frac{\pi}{10}$ is a solution.

$$(iv) \sin \frac{\pi}{5} \cos \frac{\pi}{10} = \sin \frac{2\pi}{10} \cos \frac{\pi}{10}$$

$$= 2\sin \frac{\pi}{10} \cos \frac{\pi}{10} \cos \frac{\pi}{10}$$

$$= 2\sin \frac{\pi}{10} \cos^2 \frac{\pi}{10} \quad (1)$$

$$= 2\sin \frac{\pi}{10} \left(1 - \sin^2 \frac{\pi}{10} \right)$$

$$= 2 \times \left(\frac{\sqrt{5}-1}{4} \right) \left(1 - \left(\frac{\sqrt{5}-1}{4} \right)^2 \right) \quad (1)$$

$$= \frac{\sqrt{5}-1}{2} \left(\frac{16 - (5+1-2\sqrt{5})}{16} \right)$$

$$= \frac{\sqrt{5}-1}{2} \times \frac{10+2\sqrt{5}}{16}$$

$$= \frac{\sqrt{5}-1}{2} \times \frac{5+\sqrt{5}}{8}$$

$$= \frac{5\sqrt{5}+5-5-\sqrt{5}}{16}$$

$$= \frac{4\sqrt{5}}{16} \quad (1)$$

$$= \frac{\sqrt{5}}{4}$$

COMMENTS

Many students used calculator approximations rather than properly showing this simple trig result, and lost a mark

Part (iv) not attempted by many students, only completed by (1) most able.

Question 6.

SOLUTIONS

COMMENTS.

b) Using similar triangles

radius of water surface = h .

$$\frac{dV}{dt} = \frac{dV}{dh} \frac{dh}{dt} \quad \left[\begin{array}{l} \text{vol of cone} \\ = \frac{1}{3} \pi r^2 h \\ = \frac{1}{3} \pi h^3 \end{array} \right]$$

$$24 = \pi h^2 \frac{dh}{dt} \quad (1)$$

$$\therefore \frac{dh}{dt} = \frac{24}{\pi h^2}$$

At $h = 16$ $\frac{dh}{dt} = \frac{24}{\pi \times 16^2}$

$$\frac{dh}{dt} = \frac{3}{32\pi} \text{ cm/s. } (1)$$

$$S = \pi h^2$$

$$\frac{dS}{dh} = 2\pi h$$

$$\frac{dS}{dt} = \frac{dS}{dh} \frac{dh}{dt}$$

$$\frac{dS}{dt} = 2\pi h \times \frac{dh}{dt} \quad (1)$$

At $h = 16$

$$\frac{dS}{dt} = 2\pi \times 16 \times \frac{3}{32\pi}$$

$$\frac{dS}{dt} = 3 \text{ cm}^2/\text{s. } (1)$$

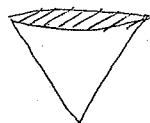
(1) students did not change $\frac{1}{3}\pi r^2 h$ to $\frac{1}{3}\pi h^3$

instead they used.

$$\frac{dV}{dr} = \frac{d(\frac{1}{3}\pi r^2 h)}{dr}$$

eventually this gave an answer $\frac{dS}{dt} = 4.5$.

(2) Students found the surface area of a cone. They should have concentrated on the surface of the area



Question 6

SOLUTIONS

COMMENTS.

a) If high tide is at 3.20pm.

Low tide would occur at 9.05 am

(6 1/4 hours earlier)

$$\frac{2\pi}{n} = \frac{25}{2}$$

$$25n = 4\pi$$

$$n = \frac{4\pi}{25}$$

$$a = 2 \quad (1)$$

$$\therefore x = -2 \cos \frac{4\pi}{25} t$$

When $x = 1$

$$10 = 9 - 2 \cos \frac{4\pi}{25} t$$

$$1 = -2 \cos \frac{4\pi}{25} t$$

$$\cos \frac{4\pi t}{25} = -\frac{1}{2}$$

$$\frac{4\pi t}{25} = \frac{2\pi}{3} \quad (1)$$

$$t = \frac{25}{6}$$

= 4 hours 10 mins

$\therefore$ Ship can safely enter at

1.15 pm. (1)

(1)

$T = 13 \text{ hrs}$

$$6.25 + 6.25 = 12.5$$

(2) students did

not calculate the first time that $x = 1$ but the second time

(2) students made $T = 6.25$

Question 6 (contd)

SOLUTIONS

COMMENTS

$$c) (i) \frac{d}{dx} \frac{1}{2} v^2 = 2 - e^{-x/2}$$

$$\frac{1}{2} v^2 = 2x + 2e^{-x/2} + C \quad (1)$$

$$x=0 \quad v=2$$

$$2 = 2 + C \therefore C = 0$$

$$v^2 = 4x + 4e^{-x/2} \quad (1)$$

$$(ii) \text{ As } x \rightarrow \infty \quad e^{-x/2} \rightarrow 0$$

$$\therefore v^2 \rightarrow 4x$$

$$v = 2\sqrt{x} \quad (1)$$

$$(iii) \text{ When } x = 100 \quad v = 20$$

$$x = 121 \quad v = 22$$

$$\therefore t = \frac{d}{s}$$

$$t = \frac{21}{21.5}$$

$$t = \frac{1}{1} \quad (1)$$

(1) Students forgot to write the constant and lost 1 mark.

Well done

Some students seem completely stumped and did not even attempt this question.

(1) For the ones that did, it was mostly well done.

Question 7.

SOLUTIONS

COMMENTS

a) (i) Since $\angle BPA = 90^\circ$ (angle in a semi-circle)

$$d^2 = BP^2 + AP^2 \quad (1)$$

$$\tan \theta = \frac{b}{PB} \quad \tan \theta = \frac{a}{PA}$$

$$\therefore d^2 = \frac{b^2}{\tan^2 \theta} + \frac{a^2}{\tan^2 \theta} \quad (1)$$

$$(ii) \tan \alpha = \frac{a}{d}, \quad \tan \beta = \frac{b}{d}$$

$$a^2 = d^2 \tan^2 \alpha, \quad b^2 = d^2 \tan^2 \beta$$

$$\therefore d^2 = \frac{d^2 \tan^2 \beta}{\tan^2 \theta} + \frac{d^2 \tan^2 \alpha}{\tan^2 \theta}$$

$$\tan^2 \theta = \tan^2 \beta + \tan^2 \alpha \quad (1)$$

b) (i) For the first particle

$$x'' = 0$$

$$x' = V \cos \theta$$

$$x = Vt \cos \theta$$

$$y'' = -g$$

$$y' = -gt + V \sin \theta$$

$$y = Vt \sin \theta - \frac{gt^2}{2}$$

Max. height occurs when $y' = 0$.

$$V \sin \theta = gt$$

$$t = \frac{V \sin \theta}{g} \quad (1)$$

Need to state this to show why you can use Pythagoras theorem.

Question 7 (contd)

$$\therefore y = V \cdot \frac{V \sin \alpha}{g} - \frac{1}{2} g \cdot \frac{V^2 \sin^2 \alpha}{g^2}$$

$$= \frac{V^2 \sin^2 \alpha}{2g}$$

$$\therefore h_1 = \frac{V^2 \sin^2 \alpha}{2g} \quad (1)$$

Similarly

$$h_2 = \frac{V^2 \sin^2 (90 - \alpha)}{2g}$$

$$h_1 + h_2 = \frac{V^2 \sin^2 \alpha}{2g} + \frac{V^2 \cos^2 \alpha}{2g} \quad (1)$$

$$= \frac{V^2}{2g}$$

Maximum range is when $y = 0$.

$$\text{ie } 0 = Vt \sin \alpha - \frac{gt^2}{2}$$

$$= t \left(V \sin \alpha - \frac{gt}{2} \right)$$

$$\therefore t = \frac{2V \sin \alpha}{g}$$

$$x = V \cdot \frac{2V \sin \alpha \cos \alpha}{g}$$

$$= \frac{V^2 \sin 2\alpha}{g} \quad (1)$$

Comments:

Many failing
to recognise
 $\sin(90 - \alpha)$
 $= \cos \alpha$.

Need to
derive these
outcomes

Question 7 (contd)

Max value of $\sin 2\alpha = 1$.

$$\therefore \max R = \frac{V^2}{g}$$

$$\frac{R}{2} = \frac{V^2}{2g} \quad (1)$$

$$= h_1 + h_2$$

(ii) Particle 1 - hits horizontal plane
when $y = 0$. Since $\tan \alpha = \frac{3}{4}$
 $\sin \alpha = \frac{3}{5}$, $\cos \alpha = \frac{4}{5}$.

$$\text{ie } 0 = 196t \times \frac{3}{5} - 9.8 \times \frac{t^2}{2} \quad (1)$$

$$= t \left(196 \times \frac{3}{5} - 4.9t \right)$$

$$\therefore t = 0 \text{ or } 24 \text{ seconds.} \quad (1)$$

Particle 2 - hits horizontal plane
when $y = 0$

$$\text{ie } y = 196t \times \frac{4}{5} - 9.8 \times \frac{t^2}{2} \quad (1)$$

$$= t \left(196 \times \frac{4}{5} - 4.9t \right)$$

$$\therefore t = 0 \text{ or } 32 \text{ s.}$$

$$\text{Time lapse} = 32 - 24$$

$$= 8 \text{ s.} \quad (1)$$

Comments:

Many
making
this more
complicated
than really is